Shakti Kumar

Physics & Mathematics Solutions Expert · PhD Scholar, IIT Kanpur

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Profile

IIT Kanpur PhD scholar (Materials Science & Engineering) with a B.Tech in Mechanical and an M.Tech in Aerospace Engineering, who writes clear, step-by-step physics and mathematics solutions that an average student can follow. Recorded 200+ worked solutions for Unacademy and taught physics, chemistry and mathematics to first-year engineering students. Solutions start from the governing principle, keep units and signs consistent, and end with a quick check. Available evenings and late at night (IST).

Subject strengths

Area	Level	Topics
Physics	Class 9–12, JEE Main & Advanced	Kinematics, laws of motion, friction, work–energy, rotation, gravitation; thermodynamics; oscillations and waves; electrostatics, current electricity, magnetism; optics and modern physics
Mathematics	Class 6–12, JEE	Algebra, trigonometry, coordinate geometry, calculus, vectors, probability
Engineering	Undergraduate, GATE	Engineering mechanics, thermodynamics, fluid mechanics, strength of materials, engineering mathematics
Chemistry	Class 9–10	Core concepts and numericals

Experience

Doubt-Solving Educator (freelance, remote) — Unacademy · Aug 2021 – Jul 2022

- Recorded 200+ step-by-step video solutions to GATE Mechanical questions in engineering mathematics, thermodynamics and fluid mechanics.

Teaching Faculty, Engineering Physics, Chemistry & Mathematics — GeeksforGurukul (ed-tech startup), PCST College, Indore · Nov 2024 – May 2025

- Taught physics, chemistry and mathematics to first-year B.Tech students through concept-based lectures and regular problem-solving sessions.

Graduate Teaching Assistant — IIT Kanpur · Jul 2023 – May 2024; Dec 2025 – present

- Supports tutorials and evaluates student work against marking rubrics for a weekly TA211 laboratory batch.

AI Training Task Author (freelance) — Alignerr (Labelbox) · 2026 – present

- Writes an engineering problem with a reference solution and an automated Python grader for training frontier AI models.

Education

- **PhD, Materials Science & Engineering (ongoing) — IIT Kanpur** · Dec 2025 – present. Coursework: Mathematics & Computational Methods; Thermodynamics of Materials; Fracture, Fatigue and Creep; Transport Phenomena (ongoing)
- **M.Tech, Aerospace Engineering — IIT Kanpur** · 2022 – 2024
- **B.Tech, Mechanical Engineering — Rajkiya Engineering College, Banda** · 2018 – 2022
- **Class XII, CBSE — Kendriya Vidyalaya No. 1, Amritsar Cantt** · 2017

Achievements

- Qualified GATE 2021 (Mechanical Engineering) in the third year of B.Tech: All India Rank 8,664 among 1,20,594 candidates (top 7.2%).
- GATE 2024 Computer Science and IT: All India Rank 22,651 among 1,23,967 candidates.
- NPTEL certifications: Engineering Mechanics – Statics and Dynamics (IIT Madras) 73%; Fluid Mechanics, Elite; Speaking Effectively (IIT Kharagpur) 90%.

Skills and availability

- **Solution writing:** handwritten step-by-step solutions, free-body diagrams, unit and dimensional checks, LaTeX typesetting
- **Languages:** English (professional), Hindi (native)
- **Availability:** evenings and late night, IST (UTC+5:30); most available after 10 PM IST